

YU YI (NICOLE) KUO

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Education

The Ohio State University

Aug 2024 - May 2026

Master of Science in Computer Science and Engineering

Columbus, OH

Coursework: Computer Architecture, Machine Learning, Parallel Computing, Data Mining, Natural Language Processing

National Ilan University

Sep 2020 - Jun 2024

Bachelor of Engineering in Computer Science and Information Engineering

Yilan, Taiwan

Coursework: Data Structure, Data Science, Operation System, Software Engineering, Algorithm

Experience

Research Assistant, The Ohio State University

Columbus, OH | Oct 2025 - May 2026

- Developed a **PyTorch** video-processing pipeline that converts high-motion facial videos into cleaner training samples through frame-by-frame motion replacement and temporal stabilization, processing **438** subject-matched video pairs
- Built a controlled evaluation workflow across **CNN**, **Transformer**, and **Mamba**-based heart-rate estimation models, testing stable, high-motion, and low-quality recordings to compare robustness across **3** model architectures
- Improved video-based heart-rate prediction accuracy by up to **67.82%** in motion-heavy settings by reducing motion artifacts while preserving subtle pulse-related facial color signals

AI Intern, NUWA Robotics

Taipei, Taiwan | May 2025 - Jun 2025

- Optimized an enterprise **RAG** pipeline by applying hybrid semantic-keyword retrieval with **BGE-M3** and **RRF** fusion, then adding cross-encoder reranking, keyword filtering, and header-based context merging to improve long-document retrieval quality and resolve production timeout issues
- Built a text-to-**MongoDB** query module with schema-aware **Chain-of-Thought** prompting, dynamic few-shot example selection, and systematic comparison across **4** prompting strategies, improving query accuracy by **13%** over baseline
- Benchmarked Cache-Augmented Generation against RAG across **4** model architectures (Qwen2-1.5B, Llama 3.2-1B/3B, Llama 3.1-8B), reducing GPU memory usage by up to **80%** and providing a technical reference for latency-sensitive architecture decisions

Auto Quality Assurance Intern, BTSE

Taipei, Taiwan | Jan 2024 - Jul 2024

- Developed **Pytest** test suites for Futures and Spot trading workflows covering liquidation triggers, position mode switching, and margin calculations, structuring manual validations into repeatable automated cases and expanding regression coverage by **56%**
- Extended existing **Jenkins** and **GitLab CI/CD** pipelines with **REST API** and **WebSocket** regression scripts as merge-triggered validation gates, replacing manual pre-deployment smoke checks and reducing release validation time by **58%**
- Correlated recurring test failures across releases by analyzing **Allure** reports and **Jira** defect patterns, surfacing root-cause clusters that reduced average defect resolution time by **27%** for a **10+**-member release team

Software Engineer Intern, Neten Digital Consultant

Yilan, Taiwan | Jul 2023 - Sep 2023

- Built a responsive **Vue.js** frontend for a venue-sharing platform with reusable component architecture, multilingual (**i18n**) support, and QR-code-driven entry flows, shipping **5** production features that increased user engagement by **29%**
- Designed **REST APIs** in **Go** with **MySQL**-backed normalized schemas and indexed lookups for hierarchical venue, user-session, and access-control data, supporting validated authentication flows that reduced average login latency by **20%**
- Architected a generic **JSON Schema** specification with multi-level tree structures to model diverse venue layouts, enabling dynamic frontend rendering from a single data format and eliminating per-venue custom page development, reducing new-venue onboarding effort by **40%**

Relevant Projects

AI-Driven Recommendation System for FDA Medical Device Safety

Nov 2025

- Built a validation pipeline for **640+** MAUDE publications using **GPT-4** MeSH enrichment, **MedCPT**, and vector search to improve medical review coverage
- Maintained at least **10** candidates under sparse metadata with exact-match plus semantic fallback retrieval, improving robustness for related-case discovery in sparse literature settings
- Validated top-**5** ranking quality with **2** raters across **60** source papers, using structured evaluation criteria to identify more relevant outputs

Breast Cancer Detection Auxiliary System with Mixed Reality

Mar 2024

- Built a breast cancer assistive diagnostic system for mammography review and achieved **70%** calcification classification accuracy with an EfficientNetB7-based model
- Benchmarked **6** CNN backbones on CBIS-DDSM crop images and improved training data quality with histogram equalization, OpenCV preprocessing, and flip/rotation augmentation
- Implemented lesion comparison and annotation workflows that surfaced **4** contour metrics on mammograms and delivered a hands-free mixed-reality interface on HoloLens 2

Technical Skills

Languages: Python, JavaScript, Go, SQL, Bash

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Vue.js, Pytest, Allure

AI/ML & Data: LLM Evaluation, RAG, Prompt Engineering, Vector Search (Milvus, Faiss), MongoDB, MySQL

DevOps & Tools: Docker, Jenkins, GitLab CI/CD, Git, Postman, Jira, REST APIs, WebSocket